Troy Anthony McMahon

CONTACT INFORMATION

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PROFESSIONAL PROFILE

Robotics researcher specializing in motion planning for manipulators, constrained systems, and high-degree-of-freedom robots. My research develops novel concepts, including Reachable Volumes and Wayfields, that combine computational geometry and sampling-based planning for grasping and manipulation. I also integrate kinodynamic planning, machine learning, and physics-based simulation to develop planning methods for mobile manipulators, industrial robots, and off-road vehicles. I have collaborated with leading researchers including Nancy Amato, Kostas Bekris, Odest Jenkins, and Ole Madsen. My work has appeared in leading robotics venues, including IJRR, ICRA, and IROS, and I have led research projects and managed collaborations across academic and industrial institutions.

RESEARCH INTERESTS

Motion Planning, Manipulation and Grasping, Kinodynamic Planning, Task and Motion Planning, Robotics, Machine Learning, Computational Geometry, Industrial Automation

EDUCATION

Ph.D., Computer Science, Texas A&M University, August 2016
- Thesis: Sampling-Based Motion Planning With Reachable Volumes
- Research Motion planning for constrained systems and high-degree-of-freedom problems
- Advisor: Nancy M. Amato

B.S., Computer Science, Physics, The University of Massachusetts, June 2005

HONORS

- *Travel Scholarship*, *IEEE International Conference on Robotics and Automation (ICRA)*, May 2015.
- *Travel Scholarship*, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, September 2014.
- *Dean's List*, Fall 2000, Spring 2001, Fall 2001, Spring 2002, Fall 2002, Spring 2003, Fall 2003, Fall 2004
- *Member of Golden Key International Honour Society*

EXPERIENCE

Aalborg University, Aalborg, Denmark

Postdoctoral Research Assistant and Project Manager **November 2022 – September 2025**

Postdoctoral Research Assistant and Project Manager for the 5G Robot Project under the mentorship of Ole Madsen. Conducted research on motion planning and visual servoing for grasping on the move in industrial environments.

Rutgers University, New Brunswick, New Jersey

Postdoctoral Research Assistant – PRACSYS Lab **January 2020 – November 2022**

Conducted research in motion planning and machine learning under the mentorship of Kostas Bekris at the PRACSYS Lab at Rutgers University. Integrated machine learning with kinodynamic planning for off-road autonomous vehicles.

Postdoctoral Research Assistant – DIMACS **October 2020 – December 2021**

Conducted research at the Center for Discrete Mathematics and Theoretical Computer Science (DIMACS) under the mentorship of Lazaros Gallos. Applied graph-based probabilistic models to study the spread of pandemics such as COVID-19.

Lecturer – Department of Computer Science **February 2020 – December 2020**
Instructor for the undergraduate Artificial Intelligence course at Rutgers University during the spring and fall semesters of 2020.

The University of Queensland, Saint Lucia, QLD, Australia

Postdoctoral Research Assistant **February 2018 – August 2019**
Conducted research in artificial intelligence and cybersecurity, with an emphasis on using planning under uncertainty for penetration testing.

Head Tutor **September 2018 – December 2018**
Head Tutor for the Artificial Intelligence course. My duties included lecturing, teaching lab and tutorial sections, and managing the other tutors for this course.

University of Michigan, Ann Arbor, Michigan

Postdoctoral Research Assistant **September 2016 – January 2018**
Conducted research in perception and manipulation under the mentorship of Odest Chadwicke Jenkins. Developed motion planning methods for manipulation with applications to task and motion planning.

Texas A&M University, College Station, Texas

Research Assistant **September 2007 – August 2016**
Conducted research in sampling-based motion planning with an emphasis on motion planning for constrained systems and high-degree-of-freedom problems. Ph.D. research introduced the concept of Reachable Volumes, which represent the regions of an environment that a robot can reach. I developed an algorithm for computing Reachable Volumes and a set of sampling methods that use Reachable Volumes to generate constraint-satisfying samples for high-degree-of-freedom problems.

Professional Activities

Mentoring: Mentored Ph.D. students Chinwe Ekenna, Sam Jacobs, Ali-Akbar Agha-Mohammadi, Daniel Tomkins, Zhefan Ye, and Aravind Sivaramakrishnan; master's students Andrew Giese, Shuvra Nath, and Dielli Hoxha; and undergraduate students Juan Burgos, Kushal Kedia, Alfred Thoft Christiansen, Elias Thomassen Dam, Julien Le Parlour, and Rudy Mpeke.

Conference and Journal Reviewer: Reviewed manuscripts for leading robotics conferences and journals including the International Conference on Robotics and Automation (ICRA), the International Workshop on the Algorithmic Foundations of Robotics (WAFR), the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Robotics: Science and Systems (RSS), the International Journal of Robotics Research (IJRR), and the Special Interest Group on Graphics and Interactive Techniques (SIGGRAPH).

Volunteer, International Conference on Robotics and Automation (ICRA), Brisbane, Queensland, Australia, 2018

Program Committee, The International Workshop on the Algorithmic Foundations of Robotics (WAFR).

Judge, REU poster competition, Dwight Look College of Engineering, Texas A&M University, College Station, TX, August, 2014

Student Volunteer, Shape Modeling International (SMI), Texas A&M University, College Station, TX, May, 2012

Student Volunteer, Workshop on Quality Software: A Festschrift for Bjarne Stroustrup, Texas A&M University, College Station, TX, April 27–28, 2012

Student Volunteer, Workshop on Intelligent Systems: A Festschrift for Richard Volz, Texas A&M University, College Station, TX, April 8–10, 2010

Future Faculty Seminar, Texas A&M University, College Station, TX, Fall 2015

Peer Reviewed Publications

Journals

[1] Troy McMahon, Aravind Sivaramakrishnan, Edgar Granados, Kostas E. Bekris, A Survey on the Integration of Machine Learning with Sampling-Based Motion Planning, Foundations and Trends in Robotics, 2022, 4(9): 266–327. doi:10.1561/2300000063

[2] Troy McMahon, Adrian Chan, Shlomo Havlin, and Lazaros K. Gallos, Spatial correlations in geographical spreading of COVID-19 in the United States, *Scientific Reports*, 699 (2022). https://doi.org/10.1038/s41598-021-04653-2

[3] Troy McMahon, Shawna Thomas, Nancy Amato. Sampling-Based motion planning with Reachable Volumes for high-degree-of-freedom manipulators. The International Journal of Robotics Research. 2018;37(7):779-817. doi:10.1177/0278364918779555

Conferences

[1] Troy McMahon, Alfred Thoft Christiansen, Elias Thomassen Dam, Casper Schou, Ole Madsen, A Two-Level Architecture for Mobile Grasping, Stuttgart Conference on Automotive Production, 2024

[2] Troy McMahon, Aravind Sivaramakrishnan, Kushal Kedia, Kostas E. Bekris, Terrain-Aware Learned Controllers for Kinodynamic Planning over Physically Simulated Terrains, 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021

[3] Aravind Sivaramakrishnan, Edgar Granados, Seth Karten, Troy McMahon, Kostas E. Bekris, Improving Kinodynamic Planners for Vehicular Navigation with Learned Goal-Reaching Controllers, 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021, pp. 9038-9043

[4] Troy McMahon, Odest Chadwicke Jenkins and Nancy Amato, Affordance Wayfields for Task and Motion Planning, 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2018, pp. 2955-2962, doi: 10.1109/IROS.2018.8594492

[5] Troy McMahon, Read Sandström, Shawna Thomas and Nancy Amato, Manipulation planning with Directed Reachable Volumes, 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2017, pp. 4026-4033, doi: 10.1109/IROS.2017.8206257

[6] Troy McMahon, Shawna Thomas and Nancy Amato, Reachable Volume RRT, 2015 IEEE International Conference on Robotics and Automation (ICRA), 2015, pp. 2977-2984, doi: 10.1109/ICRA.2015.7139607

[7] Troy McMahon, Shawna Thomas and Nancy Amato, Sampling-Based motion planning with Reachable Volumes: Application to manipulators and closed chain systems, 2014 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2014, pp. 3705-3712, doi: 10.1109/IROS.2014.6943082

[8] Troy McMahon, Shawna Thomas and Nancy Amato, Sampling-Based motion planning with Reachable Volumes: Theoretical foundations, 2014 IEEE International Conference on Robotics and Automation (ICRA), 2014, pp. 6514-6521, doi: 10.1109/ICRA.2014.6907820

[9] Troy McMahon, Sam Jacobs, Bryan Boyd, Lydia Tapia and Nancy Amato, Local randomization in neighbor selection improves PRM roadmap quality, 2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2012, pp. 4441-4448, doi: 10.1109/IROS.2012.6386061

Under Review

[1] Troy McMahon, Alfred Thoft Christiansen, Elias Thomassen Dam, Casper Schou, Ole Madsen, Visual Servoing for Grasping on the Move, *Under Review*

[2] Troy McMahon, Alfred Thoft Christiansen, Elias Thomassen Dam, Casper Schou, Ole Madsen, A Closed Loop Solution for Vision Guided Grasping on the Move, *Under Review*

Selected Workshops, Posters, and Technical Reports

[1] Troy McMahon, Kinodynamic planning with 5G for complex multi-robot interactions, Aalborg Robotics Challenge Workshop, Aalborg University, Aalborg, Denmark, September 2024

[2] Troy McMahon, Terrain-Aware Motion Planning for Physically Simulated Systems, Rutgers Robotics Workshop, Rutgers University, NJ, USA, September 2021

[3] Troy McMahon, Odest Chadwicke Jenkins, Affordance Wayfields for Task and Motion Planning, Human in-the-loop Manipulation: on the Influence of the Human Role Workshop Poster, Vancouver, Canada, September 2017

[4] Troy McMahon, Shawna Thomas, Nancy M. Amato, Directed Reachable Volumes, Late Breaking Results Poster, *IEEE International Conference on Robotics and Automation (ICRA)*, Seattle, WA, USA, May 2015

[5] Troy McMahon, Sampling-Based Motion Planning with Reachable Volumes, Ph.D. Forum Poster, *IEEE International Conference on Robotics and Automation (ICRA)*, Seattle, WA, USA, May 2015

[6] Troy McMahon, Shawna Thomas, Nancy M. Amato, Sampling-Based Motion Planning with Reachable Volumes, Technical Report TR14-003, February 2014

[7] Chinwe Ekenna, Troy McMahon, Shuvra Nath, Jory Denny, Shawna Thomas, Nancy M. Amato, A Hierarchical Neighbor Finding Strategy for Sampling-Based Motion Planning, Technical Report TR12-006, April 2012

[8] Troy McMahon, Sam Jacobs, Bryan Boyd, Lydia Tapia, Nancy M. Amato, Local Randomization in Neighbor Selection Improves PRM Roadmap Quality(Full Version), Technical Report TR12-002, January 2012

Skills

Programming: C++, Python, C, MATLAB

Robotics: ROS/ROS 2

Machine Learning:: PyTorch

Tools: CMake, Git, LaTeX, CGAL, Bullet